Henry Hsu

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SUMMARY

AI engineer whose work is mostly measurement: building the evaluators that decide whether an agent actually works. Measured attack success and privacy leakage of a red-team attack on Llama 3.1-8B, plus a defense that cut leakage 14.5pp; raised RAG retrieval precision 30% Precision@5 through reranker evaluation. Shipped production agents (MCP, tool APIs) to 20K users at MaiAgent, taking test coverage from zero to 67%. M.C.S. at Rice, graduating Dec. 2026, available Feb. 2027.

WORK EXPERIENCE

Software Engineering Intern
Jun. 2026 – Aug. 2026

Google *Taipei, Taiwan*

- **Integration Test Infrastructure:** Built **3** test infrastructures from a **zero** baseline (Java, Python), onboarding **400+** integration test suites across **200+** smart home device types, with **90%+** unit test coverage for the infrastructure itself
- **Agentic Failure Triage:** Developed an LLM agent that triages scheduled integration test runs and surfaces root-cause summaries directly on issue tickets, replacing manual log inspection and cutting failure narrow-down time by **20+ minutes** per issue
- **Local Device Reproduction:** Extended the agent to reproduce device failures locally against virtual devices instead of waiting on cloud runs, cutting manual reproduction time by **85%**; **shipped and adopted by the team**

Generative AI Team Lead
Dec. 2024 – Aug. 2025

MaiAgent Co., Ltd. *– Award-Winning B2B GenAI Startup* *Taipei, Taiwan*

- **Building AI Agents:** Upgraded chatbots into AI agents by integrating memory systems, tool APIs, and MCP Client, resulting in **120%** partner growth (CTBC Bank, MSI, HPE, iGroup) and user expansion from **3K** to **20K** users while cutting LLM token usage by **67%+**
- **Contributing to Open Source:** Contributed **14** merged pull requests to the **LlamaIndex** open-source project (**15K+ stars**), fixing bugs and adding features in integrations with AWS Bedrock, Claude, Elasticsearch, Cohere, OpenAI, MCP Client, and AI Agent Workflow
- **Testing Framework:** Introduced pytest testing framework for unit and E2E coverage to prevent production regressions, achieving **67%** test coverage from **zero** baseline, which reduced production hotfixes by **90%** initially and sustaining a **50%** reduction long term
- **Technical Roadmap & Leadership:** Defined GenAI product roadmap for a 10-person startup; led feasibility evaluation and end-to-end delivery of production features including Agentic RAG, Artifact Generation, Tagging & Permission System, and Information Retrieval

Generative AI Intern
Sep. 2024 – Dec. 2024

MaiAgent Co., Ltd. *– Award-Winning B2B GenAI Startup* *Taipei, Taiwan*

- **Pipeline Scaling:** Refactored with async framework and dual-database synchronization (RDB/Vector DB), achieving **3.5x** faster parsing and scaling from **3M** to **20M+** text chunks while **doubling** indexing performance
- **Reranker Integration:** Integrated Cohere and BGE open-source rerankers as a customer-selectable retrieval feature, improving RAG precision on large documents by **30% Precision@5** and **7% Precision@10**

EDUCATION

Rice University - Top 20 U.S. University
Aug. 2025 – Dec. 2026 (Expected)

M.C.S., Computer Science (GPA: 4.00/4.00) *Houston, TX*

National Yang Ming Chiao Tung University (NYCU) - Top 3 Taiwan University
Sep. 2020 – Jun. 2024

B.S., Industrial Engineering and Management (GPA: 4.07/4.30, Two-Time Dean's List) *Hsinchu, Taiwan*

Minor: Computer Science (Domain GPA: 4.13/4.30)

PROJECTS

Defending LLM Agents Against Tool Schema Poisoning | *PyTorch, Llama 3.1, Activation Steering, Slurm* **Apr. 2026 – May 2026**

- **Threat Model:** Reproduced a tool-schema poisoning attack where deceptive parameter descriptions make structured-output agents auto-fill private data, setting a **56%** attack-success and **52%** privacy-leak baseline on Llama 3.1-8B
- **Representation-Level Defense:** Probed hidden states to locate a linear safety direction (**96%** probe accuracy at layer 10) and steered activations along it at inference, cutting privacy leakage **14.5pp** with no loss in task success
- **Defense Comparison:** Benchmarked steering against an in-context reflection wrapper in single-turn runs: both cut leakage about 14pp, but the wrapper raised attack success **6pp** and lowered task success 6pp, while steering kept task success

AWARDS

Presidential Hackathon Winner (2024, Top 5 Nationally) - Urban noise agent: structured-output LLM pipeline and tool calling

23rd Golden Peak Award (2025) - Outstanding Commercial Product, MaiAgent AI Platform

SKILLS

Languages: Python, Java, C/C++, TypeScript | **GenAI:** LlamaIndex, LangChain, LangGraph, MCP | **Backend:** Django, FastAPI
Evaluation: RAGAS, Precision@k, MAR@k, Attack-Success & Leak-Rate Evaluators, Factorial Ablations, Benchmark Harnesses, pytest
ML: PyTorch, numpy, pandas, scikit-learn | **Data:** PostgreSQL, Milvus, Elasticsearch | **Infra:** Docker, git, Slurm, AWS Bedrock/EC2/S3